
BRAIN TUMOR TYPE CLASSIFICATION

DEEP LEARNING FOR VISUAL RECOGNITION



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ABSTRACT

Accurate classification of brain tumors from MRI scans is critical for determining appropriate treatment strategies, yet it remains challenging due to the biological heterogeneity of tumor types. This study evaluates two deep learning approaches for the classification of four categories: glioma, meningioma, pituitary adenoma, and no tumor. First, we employed a transfer learning strategy using state-of-the-art architectures (Swin Transformer, ResNeXt, MobileNet) to establish baseline performance. Second, we developed a Hierarchical Multi-Task Learning (MTL) U-Net that simultaneously performs image reconstruction, binary tumor detection, and subtype classification, utilizing an adaptive Exponential Moving Average (EMA) loss weighting scheme to balance competing objectives.

Initial results on the raw dataset showed high validation accuracy (>98%) but a significant drop in testing accuracy (77%), primarily driven by poor recall in the glioma class across all models. Latent space analysis using UMAP revealed a substantial distributional shift between validation and test samples for gliomas, suggesting a dataset-level inconsistency rather than architectural failure. Following the removal of 1,113 duplicate images and data curation, the model's testing accuracy improved to 97.3%. Our findings demonstrate that while complex multi-task architectures can learn robust anatomical features, rigorous data curation and latent space inspection are paramount for ensuring model generalization in medical imaging.

Keywords Classification · Transfer Learning · Multi Task Learning

1 Introduction

Accurate tumor classification is a critical step with substantial implications for patient treatment and follow-up. Brain neoplasms such as meningiomas, pituitary adenomas, low-grade gliomas, and glioblastomas differ markedly in their biological behavior, aggressive growth dynamics, and infiltration patterns, necessitating distinct treatment strategies.

In Denmark, approximately 1,600 patients are diagnosed with primary brain tumors annually. Of these, meningiomas account for approximately 36%, gliomas for 38%, pituitary adenomas for 6-8%, and the remaining 20% include schwannomas, ependymomas, and other less prevalent tumors [1, 2].

The diagnostic process typically begins when a patient presents with seizures or headaches and is referred for an MRI scan. The scan is reviewed by a radiologist who offers a differential diagnosis—identifying the most probable condition while listing alternative candidates. The final diagnosis is established through a process of elimination, where histopathology combined with molecular markers has become the definitive standard following the WHO 2021 guidelines [3]. However, in specific cases involving deep-seated tumors or those located near eloquent brain areas, obtaining pathology samples is considered too risky. In these scenarios, the diagnosis—and subsequent treatment plan—must rely solely on non-invasive MRI information.

*Implementation details for the work included here can be found at https://github.com/project2025/DL_course_

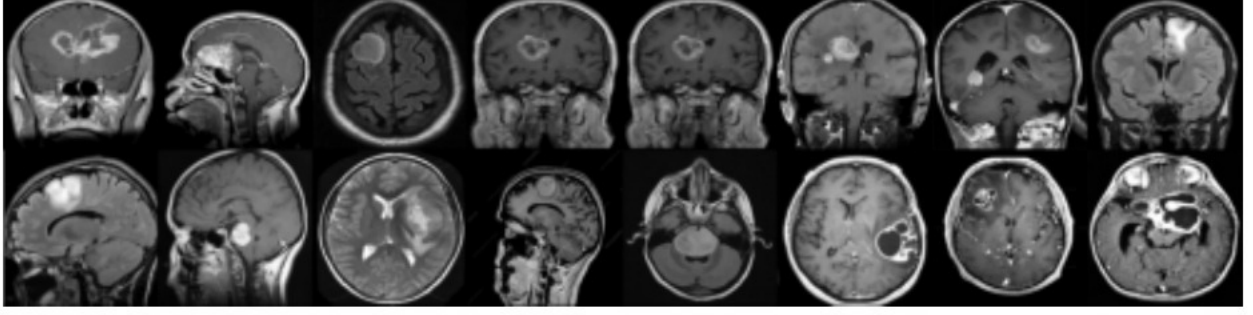


Figure 1: Sample images of Dataset1 testset

Learning the relevant imaging features that best distinguishes the different tumor types using cases where all relevant clinical information (ground truth) is available may thus improve the diagnosis and treatment of the patients whose available information is limited to MRI only. The aim of this study in there for to:

- Apply transfer learning to classify patient MRI scans into four classes: meningiom, glioma, pituitary and non tumors
- Design a multitask convolutional neural network to perform the same task.

1.1 Related Work

In the design of our network, we employ the U-Net backbone introduced by Ronneberger et al. [4]. Unlike standard classifiers, we retain the decoder to perform image reconstruction, ensuring the encoder captures robust global anatomical features. To manage the competing objectives of reconstruction and classification, we draw on Multi-Task Learning principles discussed by Kendall et al. [5]. However, instead of uncertainty-based weighting, we implemented an adaptive Exponential Moving Average (EMA) normalization strategy to dynamically balance loss magnitudes during training.

2 Materials and Methods

2.1 Data

The data used for training was retrieved from 2 different sources. Both dataset contained brain MRI images of meningiomas, pituitary adenomas, gliomas and non tumor cases. The input images were all (512x512)x3 channels already normalized to the range [0,1], thus no additional normalization was performed.

Dataset 1 was sourced from <https://www.kaggle.com/datasets/sartajbhuvaji/brain-tumor-classification-mri/data> Figure 1 shows example images of the test set.

Dataset 2 extended dataset 1 with additional images from other sources (<https://www.kaggle.com/datasets/rishantenis/brain-tumor-mri-dataset/data>)

From Figure 1 it became apparent that there was a number of image duplicates in the different dataset (center images in top row). In order to avoid inflated performance estimates, which can seriously undermine the validity of model evaluation. Dataset2 was curated to remove images that occurred more than once. This resulted in three dataset with class numbers as in figure 2

2.2 Methods

2.2.1 Data Preprocessing and Augmentation:

To ensure input consistency across the different data sources and to improve model generalization, we implemented a transformation strategy utilizing the torchvision library. Images were converted to single-channel grayscale tensors to remove redundant channel information and resized to spatial dimensions of 512×512 pixels with antialiasing to preserve anatomical detail. To mitigate the risk of overfitting and simulate the variability inherent in clinical acquisition (e.g., patient positioning and scanner differences), a data augmentation pipeline was applied exclusively to the training subset. This pipeline included:

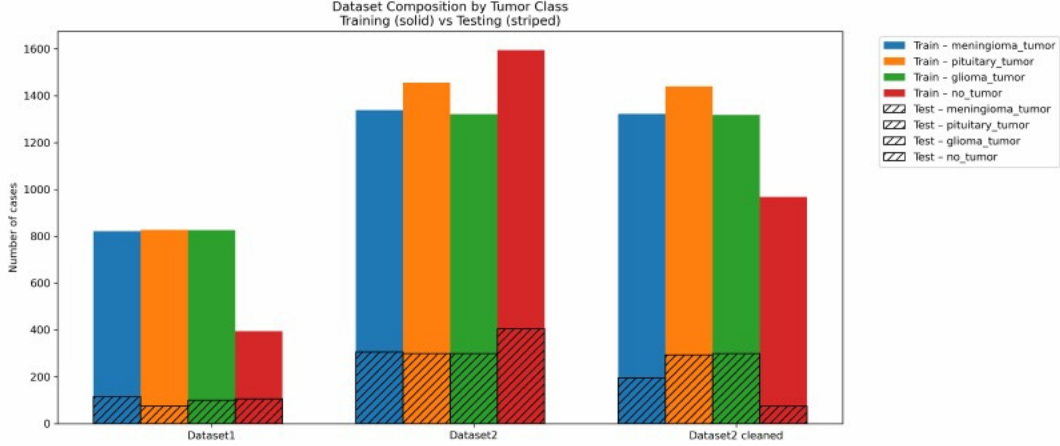


Figure 2: Dataset class distribution for the different datasets.

- **Geometric Transformations:** Random affine transformations allowing for rotations ($\pm 10^\circ$), translations (up to 2%), and scaling factors (0.95–1.05).
- **Intensity Transformations:** ColorJitter was applied to vary brightness and contrast by a factor of 0.10.
- **Artifact Simulation:** A Gaussian blur (kernel size 3) was applied with a probability of $p = 0.2$ to mimic motion artifacts or lower-resolution scans.

The validation and test sets remained unaugmented to serve as stable benchmarks for performance evaluation.

2.2.2 Dataset Partitioning

The cleaned dataset was loaded using the standard ImageFolder structure. While the test set was maintained as a distinct entity to prevent data leakage, the remaining data was partitioned into training and validation subsets using a stratified hold-out strategy. We allocated 20% of the training data for validation purposes.

2.2.3 Class Imbalance Mitigation

Recognizing the potential for class imbalance to bias the model towards the majority class (e.g., Glioma), we implemented a weighted sampling strategy. We extracted the target labels from the training Subset and calculated class-specific weights (W_c) based on the inverse frequency of each class, such that:

$$W_c = \frac{1}{N_c}$$

where N_c represents the number of samples for class c . These weights were utilized to initialize a WeightedRandomSampler with replacement. This ensures that during training, batches are constructed to approximate a uniform distribution of classes, preventing the optimization process from being dominated by the most prevalent tumor types.

2.3 Model architectures

Training was conducted using two distinct approaches: an initial transfer-learning approach to identify preprocessing, reasonable hyperparameters and baseline performance, followed by a hierarchical multi-task learning (MTL) approach. Distinct optimization objectives were formulated for each phase.

2.3.1 Transfer Learning (Warm-Start): Optimization, Loss and Networks

To initialize the feature encoder and benchmark baseline performance, we first optimized the network as a single-task classifier. We employed the AdamW optimizer with a standard Cross-Entropy objective. To improve model calibration and prevent overfitting, we utilized label smoothing ($\epsilon_{ls} = 0.05$). For an input sample with class label $y \in \{0, 1, 2, 3\}$ and unnormalized logits $\mathbf{z}$, the loss was defined as:

$$\mathcal{L}_{CE} = \text{CrossEntropy}(\mathbf{z}, y; \epsilon_{ls} = 0.05)$$

Cosine Annealing learning rate schedule with an initial learning rate of $3e-4$ was used through out this work.

A number of network candidates were used in a transfer learning scheme: VGG16 (vgg16); ResNext (ResNeXt50_32x4d); SwinTransformer (swin_b); Vision Transformer (vit_l_32); MobileNetV2 (mobilenet_v2); MobileNetV3 (mobilenet_v3)

Each pretrained model was finetuned to Dataset 1 i.e. unfreezing all pretrained weights for the ImageNet-1K dataset and allowed to run for up to 500 epochs. The classification head was replaced to adapt to the tumour-type prediction task.

2.3.2 Hierarchical Multi-Task Learning

In the second approach, a UNet backbone similar to that of Ronneberger et al. [4] was extended to perform multitasking similar to [5] of three joint tasks: image reconstruction (self-supervision), binary tumor detection, and specific subtype classification.

Task-Specific Loss Components

- **Reconstruction Loss ($\mathcal{L}_{recon}$):** To encourage the encoding of robust, global anatomical features, we minimized the Mean Squared Error (MSE) between the input MRI slice x and the autoencoder reconstruction $\hat{x}$:

$$\mathcal{L}_{recon} = \|\hat{x} - x\|_2^2$$

- **Binary Detection Loss ($\mathcal{L}_{bin}$):** A binary head predicted the presence or absence of a tumor, optimized via Binary Cross-Entropy with logits:

$$\mathcal{L}_{bin} = BCEWithLogits(z_{bin}, y_{bin})$$

- **Subtype Classification Loss ($\mathcal{L}_{sub}$):** To accommodate the hierarchical label structure, this loss was masked. It was computed only for samples where a tumor was present ($y_{sub} \neq -1$), ensuring the model was not penalized for failing to classify the subtype of healthy tissue:

$$\mathcal{L}_{sub} = CrossEntropy(z_{sub}, y_{sub}), \quad \forall i \in \mathcal{I}_{tumor}$$

Adaptive Loss Balancing The distinct tasks (reconstruction vs. classification) operate on different scales, posing a risk that one objective might dominate the gradient updates. To mitigate this, we employed an adaptive normalization strategy based on the Exponential Moving Average (EMA) of historical loss magnitudes. The hierarchical classification loss was first aggregated:

$$\mathcal{L}_{cls} = \mathcal{L}_{bin} + \alpha \mathcal{L}_{sub}$$

where α balances the subtype contribution. Subsequently, both the reconstruction and classification terms were dynamically scaled by their respective running averages ($\hat{m}$):

$$\tilde{\mathcal{L}}_{recon} = \frac{\mathcal{L}_{recon}}{\hat{m}_{recon} + \varepsilon}, \quad \tilde{\mathcal{L}}_{cls} = \frac{\mathcal{L}_{cls}}{\hat{m}_{cls} + \varepsilon}$$

The final multi-task objective minimized during training was the sum of these stabilized terms:

$$\mathcal{L}_{total} = \tilde{\mathcal{L}}_{recon} + \tilde{\mathcal{L}}_{cls}$$

This approach ensures a stable trade-off between semantic interpretation (classification) and feature preservation (reconstruction) throughout the optimization trajectory.

Hierarchical Multi-Task U-Net design details: To enable simultaneous image reconstruction and hierarchical tumor classification, we developed a Multi-Task Learning (MTL) architecture based on the classical U-Net backbone. The network consists of a shared encoder, a reconstruction decoder, and a dual-head classification branch.

Shared Encoder: The encoder captures hierarchical features at increasing levels of abstraction. It consists of four downsampling blocks. Each block is composed of:

- **Stacked Convolution:** Two 3×3 convolutional layers with padding (to maintain spatial dimensions), each followed by Batch Normalization and a ReLU activation function.
- **Pooling:** A 2×2 Max Pooling operation with stride 2 to reduce spatial resolution.

The feature channel progression is symmetric, doubling at each depth level ($64 \rightarrow 128 \rightarrow 256 \rightarrow 512$), culminating in a bottleneck representation with 1024 channels

Classification Branch (The Multi-Task Head): At the bottleneck (the deepest layer of the network), the high-level feature maps represent the semantic content of the image. To repurpose these features for classification without destroying spatial information needed for reconstruction, we implemented a branching strategy:

- **Feature Aggregation:** The bottleneck features are compressed using Global Average Pooling (GAP) to generate a spatially invariant feature vector.
- **Latent Projection:** This vector is flattened and passed through a shared dense layer (Linear $\rightarrow$ ReLU) to project it into a latent space of size 256.
- **Hierarchical Heads:** Two parallel fully connected layers branch from this shared latent vector:
 - **Binary Head:** Outputs a single logit representing the probability of "Tumor" vs. "No Tumor."
 - **Subtype Head:** Outputs $N = 3$ logits corresponding to the specific tumor classes (Glioma, Meningioma, Pituitary).
 - **Regularization:** Both heads utilize Dropout ($p = 0.2$) to prevent overfitting on the dense layers.

Reconstruction Decoder: Parallel to the classification branch, the decoder reconstructs the original input image from the bottleneck features. It employs:

- **Upsampling:** 2×2 Transposed Convolutions to restore spatial resolution.
- **Skip Connections:** Features from the corresponding encoder level are concatenated with the upsampled features to recover fine-grained spatial details lost during pooling.
- **Output:** A final 1×1 convolution maps the feature vectors back to the original single-channel image space for the reconstruction loss calculation.

2.4 Latent Space Analysis and Label Error Detection

To validate the semantic separability of the learned features and identify potential dataset inconsistencies, we analyzed the latent feature space of the trained MTL model. We extracted the 256-dimensional latent vectors ($z \in \mathbb{R}^{256}$) from the shared projection layer of the classification head for all validation and test samples. To visualize this high-dimensional space, we employed Uniform Manifold Approximation and Projection (UMAP), a non-linear dimensionality reduction technique. UMAP was configured to project the feature vectors into a two-dimensional manifold using a Euclidean distance metric, preserving both local cluster structure and global data geometry. This visualization served as a primary diagnostic tool for label noise identification. We operated on the premise that a highly converged encoder groups images based on visual content; therefore, samples labeled as class A but embedded deeply within a dense cluster of class B were flagged as potential annotation errors or data leakage artifacts rather than model misclassifications.

3 Results

3.1 Transfer learning

Transfer learning was performed and performance was tested on Dataset 1 using above pretrained models and hyperparameters. Figure 3 is an example of the validation loss and validation accuracy for the different models indicating little to no overfitting across all models. Summary of the different models accuracies for validation and test set is summarized in Table 1.

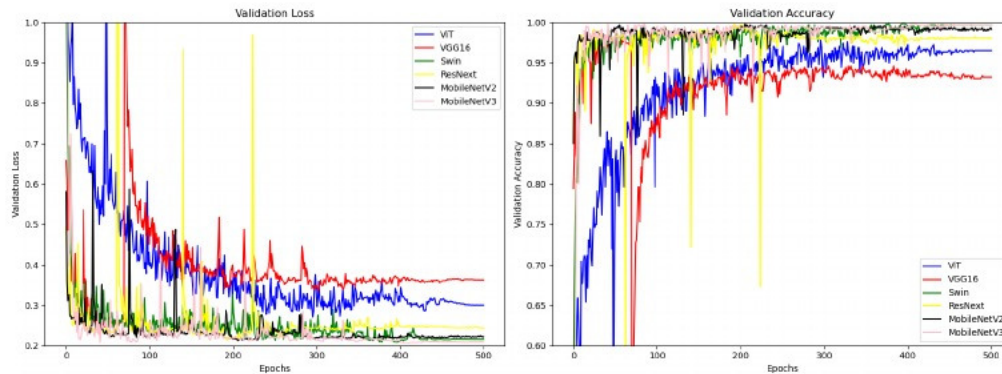


Figure 3: Validation loss/accuracy across models.

Table 1: Classification results for the validation and test set

Model Name	Validation Accuracy	Test Accuracy
VGG16	0.9862	0.7990
ResNext	0.9983	0.8052
Swin Transformer	1.0000	0.8038
Vision Transformer	0.9983	0.8038
MobileNetV2	0.9983	0.8254
MobileNetV3	0.9983	0.7895

The Swin Transformer network showed the best validation accuracy however MobileNetV2 generalized better to the test set. The confusion matrix for Swin Transformer is found in Table 2. The overall performance of the model on the independent test set was dominated by substantial class-specific variation. While meningioma, pituitary tumors, and no-tumor cases were classified with high recall (1.00, 0.80, and 1.00, respectively), the model showed marked difficulty generalizing to the glioma class. Only one third of glioma cases were correctly identified (recall = 0.33), with the remainder predominantly misclassified as meningioma or no-tumor. This pattern indicates that the feature representation learned during training does not transfer adequately to the glioma samples in the test set, suggesting a distributional mismatch or increased heterogeneity within this class. The confusion matrix further highlights that misclassification errors are largely driven by this single class, whereas the remaining tumor categories exhibit stable and well-separated decision boundaries.

Table 2: Confusion matrix for the four-class tumor Swin Transformer classification model.

True / Predicted	Glioma	Meningioma	No tumor	Pituitary
Glioma	33	39	26	2
Meningioma	0	115	0	0
No tumor	0	0	129	0
Pituitary	0	9	6	59

Finally, the ability of the network to identify relevant image regions was evaluated using Gradient-weighted Class Activation Mapping (Grad-CAM) as shown in figure 4. This suggests the network has learned the relevant imaging features despite the poorer performance on the testset.

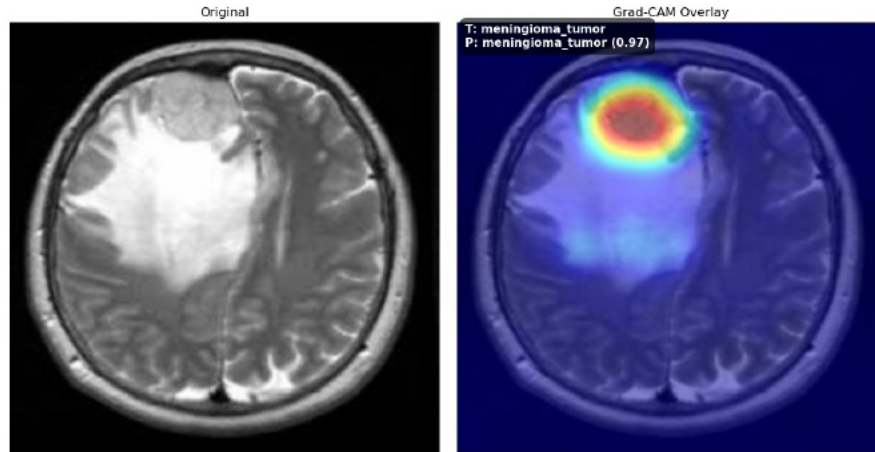


Figure 4: Grad-CAM visualization illustrating network attention for a meningioma case with 97% prediction confidence.

3.2 Multi Task Learning Approach

3.2.1 Dataset 1

For the first dataset the model training performed as in figure 5 with a similar testset accuracy as the transfer learning experiments of 0.7775. On the left is the total training and validation loss while the accuracy is divided into the different loss function elements: Training and validation accuracy for the 4 classes, validation accuracy for tumor/notumor classification and finally, the 3 tumor sub classification accuracy.

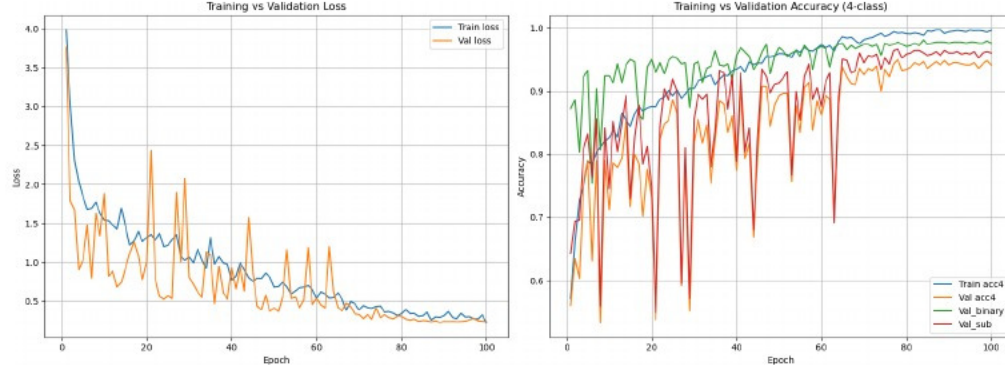


Figure 5: Multi-Task Loss and accuracy.

Table 3: Confusion matrix for the tumor classification

True / Predicted	Glioma	Meningioma	No tumor	Pituitary
Glioma	25	28	37	10
Meningioma	0	109	5	1
No tumor	0	0	129	0
Pituitary	0	5	7	62

When compared with the confusion patterns observed in the transfer learning models (table 2), the Multi task network exhibits a similar failure mode centred on the glioma class (table 3). In the transfer learning setting, gliomas accounted for the majority of misclassifications, primarily being assigned to meningioma or no-tumor. This trend persists in the multi task model, where glioma recall remains low and errors are again distributed across these two classes. By contrast, classes that were well separated in the transfer learning experiments—particularly no-tumor and meningioma—retain high recall and stable decision boundaries in the hierarchical approach. The strong correspondence between both confusion matrices indicates that the underlying limitation is not model-specific but reflects a systematic mismatch between the glioma samples in the training and test sets, rather than an architectural shortcoming of the hierarchical framework. To investigate this further the common latent space was visualized using the UMAP representation as seen in figure 6 for both validation and test set indicating that there indeed is a difference of especially the glioma cases in the test set (blue crosses) compared to the validation set (blue circles).

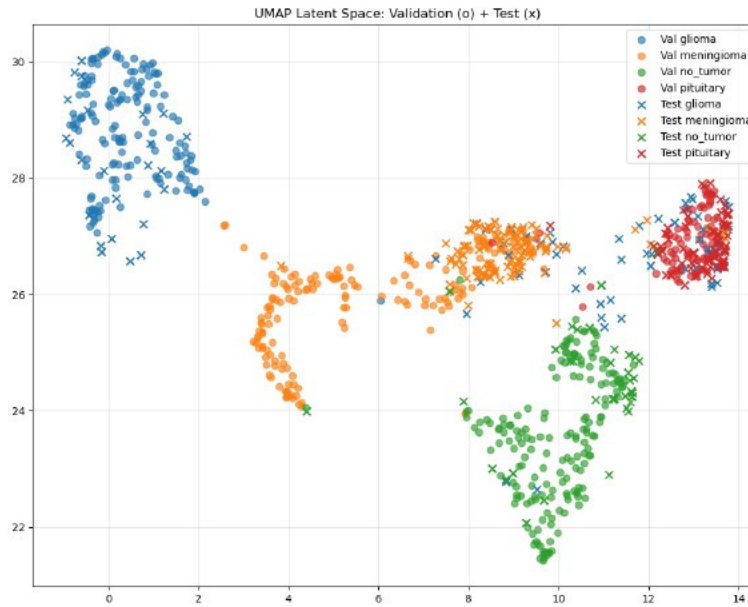


Figure 6: UMAP representation of the different classes in both validation and test set.

3.2.2 Dataset 2 (raw and cleaned)

For the second dataset the model training performed similar to Dataset1 and for the accuracy of the raw Dataset2 testset the classification accuracy was 0.7747. After removing 1113 duplicates from Dataset 2 (->Dataset2 clean) the testset accuracy improved to a staggering 0.9733.

4 Discussion

4.1 Performance Analysis and the "Glioma Gap"

The primary objective of this study was to compare standard transfer learning classifiers against a custom Multi-Task Learning (MTL) framework. In the initial experiments on Dataset 1, both approaches exhibited a distinct performance discrepancy. While the models achieved near-perfect validation accuracy, the testing accuracy plateaued around 78-82%. The confusion matrices for both the Swin Transformer (Transfer Learning) and the MTL U-Net revealed that this error was almost exclusively driven by the misclassification of gliomas, which were frequently confused with "no tumor" or meningioma cases.

The consistency of this error pattern across completely different architectures (ViT vs. CNN-based U-Net) strongly suggested a data-centric issue rather than a model capacity limitation. This hypothesis was confirmed via UMAP analysis, which visualized the latent feature space. The projection showed that glioma samples in the test set formed a cluster distinct from those in the validation set, indicating a domain shift or distributional mismatch likely caused by the sourcing of the dataset.

4.2 Efficacy of Adaptive Multi-Task Weighting

A core technical challenge in Multi-Task Learning is preventing task dominance, where a loss with a larger magnitude (e.g., Cross-Entropy) overshadows a smaller loss (e.g., MSE). The implemented adaptive weighting strategy, based on the Exponential Moving Average (EMA) of historical losses, successfully mitigated this issue. By dynamically normalizing the reconstruction and classification losses to a common scale, the model was able to converge on both tasks simultaneously without manual hyperparameter tuning. This suggests that dynamic loss balancing is a viable alternative to static weighting in medical imaging tasks where objective scales may vary unpredictably.

4.3 The Impact of Data Curation

Perhaps the most significant finding of this work lies in the impact of data quality control. The initial dataset contained substantial redundancy, which likely inflated validation scores and masked generalization issues. The removal of 1,113 duplicate images (Dataset 2 Cleaned) resulted in a massive performance jump, raising the test accuracy from 77% to 97.33%. This underscores a critical lesson in medical AI: algorithmic sophistication cannot compensate for compromised data integrity. The high performance of the cleaned dataset validates that the underlying models were capable of learning distinct tumor features once the training and testing distributions were aligned.

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